

Operational Shadows of Hilbert-Space Probabilities

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At one frozen setting, the probabilities observed in a sharp quantum context are indistinguishable, as detector-click statistics, from ordinary probabilities on the atoms of a classical partition. But an actual analyzer usually comes with a calibrated knob. If the measurement configuration is co-varied continuously through this physical parameter, the operational object is no longer one point of a simplex but a response curve. Classical linear responses, Malus-type Hilbert-space responses, parameter cheats, softmax links, and Heaviside limits are different ways of turning a knob setting into probabilities. Continuity, calibration, and preservation of the physical composition law are then part of the experimental meaning of the knob. This difference can already be visible before any multi-context obstruction is considered. The static operational coincidence can also persist for two intertwined contexts: if the common outcomes receive the same probabilities, the remaining masses can always be coupled by a classical joint distribution. Genuine multi-context nonclassicality begins when a family of local shadows cannot be glued into one nonnegative global distribution or one simplex factorization. Farkas’ lemma gives the exact alternative: either the classical extension exists, or a separating linear inequality certifies its impossibility.

Informal introduction.— A hand held before a lamp casts a flat shadow on a wall. Many different hands, postures, and objects can cast the same silhouette. If the wall is the only thing one is allowed to inspect, the shadow is all one has; but it should not be mistaken for the object that produced it. This is the old warning of Plato’s cave: the prisoners take the passing shadows for reality. The image contains a second warning that is just as relevant here. The one who turns around and returns with an account of the objects and the light behind the shadows is not automatically believed; he may appear blinded or confused, and the cave’s inhabitants may punish the attempt to free them from their own evidence [1, Book VII, 514a–517a]. Thus the metaphor cuts both ways. A shadow should not be reified as the thing itself, but neither is the hidden machinery behind it given by the shadow alone. To speak of an ontology casting the shadow is already to add a reconstruction, a model of what lies off the wall.

Something similar happens in a single sharp measurement. A system is sent into a detector bank, one detector clicks, and after many repetitions one has a list of relative frequencies. This list is a point of a probability simplex. Looking only at that list, one cannot tell whether it was generated by a classical partition, by squared Hilbert-space amplitudes, or even by a more exotic admissible weight. The generating geometries are different, but a single frozen setting can flatten them into the same operational shadow.

The missing character is the knob: a tangible handle on the apparatus. A polarizer rotates, a Stern–Gerlach magnet tilts, a phase is advanced, a temperature or score is dialed. Turning such a knob is itself part of the experiment: equal increments can be calibrated, small changes can be made continuously, and successive rotations compose. Thus the knob is not merely a coordinate on an

interval; it carries a physical action. Once this physical variation is retained, the shadow is no longer a single point but a trajectory through the simplex. A classical model may hit any one point of a quantum response curve, but it need not reproduce the whole curve with the same symmetry-preserving knob. This is the intuition behind the “parameter cheats”: they work by deforming the scale on which the knob is read, thereby changing the physical action represented by equal turns of the apparatus.

The paper follows this stratification. One frozen context gives a shadow. One calibrated variation gives a response law. Two weakly intertwined contexts are still classically couplable when their shared atoms agree. A rich enough web of contexts can fail to have any global nonnegative extension, and Farkas’ lemma gives the exact alternative: either the classical extension exists or a separating inequality must appear.

Operational shadow.— Let $C = \{P_1, \dots, P_d\}$ be a maximal rank-one projective measurement on a d -dimensional Hilbert space. In a run of the experiment only one detector clicks, and the operational data associated with this single context are the frequencies

$$p_i \geq 0, \quad \sum_{i=1}^d p_i = 1. \quad (1)$$

They are the same kind of object as a probability distribution on the atoms of a classical partition. The phrase “same operational shadow”¹ will mean this: the observable click statistics in the context are identical, even if

¹ This use of “shadow” is the Plato-cave metaphor used here, not the randomized measurement technique known in quantum information as “classical shadows”.

the underlying state spaces used to generate them are different.

The difference between the two theories is already present in the generating maps. A classical state on a d -atom partition is a point of the simplex Δ_{d-1} , with pure states at the vertices and mixtures represented affinely. A pure quantum state is a ray in complex projective space, $[\psi] \in \mathbb{CP}^{d-1}$, and Born's rule gives

$$s_C([\psi]) = (\langle \psi | P_1 | \psi \rangle, \dots, \langle \psi | P_d | \psi \rangle). \quad (2)$$

In an eigenbasis $P_i = |e_i\rangle\langle e_i|$, this is the squared projection map $p_i = |\langle e_i | \psi \rangle|^2$. Thus the same probability simplex is reached through a different geometry: affine weights in the classical case, squared Hilbert-space amplitudes in the quantum case.

Proposition 1. *For one maximal rank-one context C , the pure-state shadow map $s_C : \mathbb{CP}^{d-1} \rightarrow \Delta_{d-1}$ is onto.*

Proof. Given $p = (p_1, \dots, p_d) \in \Delta_{d-1}$, choose arbitrary phases ϕ_i and set

$$|\psi\rangle = \sum_{i=1}^d \sqrt{p_i} e^{i\phi_i} |e_i\rangle. \quad (3)$$

Then $\langle \psi | P_i | \psi \rangle = p_i$. The phases, modulo a global phase, are invisible to this context. Hence a single context sees the full ordinary probability simplex and nothing beyond it. $\square$

This proposition is deliberately modest. It does not say that the classical and quantum state spaces are isomorphic. It says only that after forgetting everything except one detector bank's click frequencies, both theories cast the same operational shadow. The phrase "forgetting" is essential: it freezes the apparatus setting and discards the way in which the probabilities change when the apparatus is moved.

Response functions and knobs.— In the laboratory the setting is rarely just an abstract label. A polarizer is rotated, a Stern–Gerlach magnet is tilted, a phase is advanced, or a score parameter is changed. The detector bank may still be the same operational device, but the measurement configuration is co-varied with a physical parameter. Once this knob is retained, the observed data are not merely $p \in \Delta_{d-1}$, but the curve swept out by p as the knob is turned. If K denotes the setting space of the knob, the relevant object is the response function

$$R_C : K \longrightarrow \Delta_{d-1}, \quad k \longmapsto (p_1(k), \dots, p_d(k)). \quad (4)$$

A classical partition can hit any one point of this curve. That is not the same as reproducing the curve with the same physical knob. A calibrated angle is not just any coordinate on an interval. It has a topology, an isotropy convention, and a composition law: two successive rotations by θ_1 and θ_2 compose as $\theta_1 + \theta_2$, modulo the

relevant period. More generally, a calibrated knob is a setting space K carrying the action of a physically implemented transformation group G , often a homogeneous space G/H . Admissible changes of coordinate should preserve this action, or at least intertwine it with the corresponding physical action on the apparatus. Such calibration can be checked independently of the output probabilities, for instance by composing equal rotations, verifying return after a full period, or using an interferometric phase reference. A harmless reparameterization must therefore intertwine this physical action; for a rotation knob it must preserve the operational meaning of equal turns and composed turns, up to the convention used to orient or scale the dial. Continuity is part of this structure, but it is not the whole structure: a continuous deformation of the coordinate can still fail to preserve the group action. These facts are operational, because equal angular increments, composed rotations, and continuous sweeps can be implemented and compared.

One may object that, once the knob is turned, one has already left the original context and entered a many-context situation. Formally this can be a useful description: each fixed angle may be represented by its own maximal context. But that is not the emphasis here. A knob sweep is not primarily a finite list of unrelated configurations to be glued by a marginal-extension test. It is a functional response to an argument, such as $\theta \mapsto p(\theta)$, where the argument has a physical scale, continuity, and composition or symmetry law. The question is therefore not only which static contexts are present, but how the probabilities vary when the apparatus is continuously moved.

This is already enough to distinguish response laws in a single continuously varied context. A classical model of correlated directions gives a linear equal-outcome response, whereas a spin-1/2 singlet gives the curved Malus-type response. For these binary examples write $R(\theta) = P^\pm(\theta)$. On $0 \leq \theta \leq \pi$ one may write

$$\begin{aligned} P_{\text{cl}}^\pm(\theta) &= \frac{\theta}{\pi}, & P_{\text{qm}}^\pm(\theta) &= \sin^2 \frac{\theta}{2}, \\ P_s^\pm(\theta) &= H\left(\frac{2\theta}{\pi} - 1\right), \end{aligned} \quad (5)$$

where the last, Heaviside-type response is the discontinuous stronger-than-quantum limit [2]. At a single value of θ , each line only gives an ordinary two-outcome probability distribution. As θ is co-varied, however, the variational response is completely different: linear, smoothly curved, or step-like.

The 2001 "parameter cheat" construction makes this point particularly transparent. If one introduces a new parameter δ by $\theta = \pi \sin^2(\delta/2)$, then the classical expression $P_{\text{cl}}^\pm(\theta) = \theta/\pi$ can be made to look quantum: $P_{\text{cl}}^\pm(\theta(\delta)) = \sin^2(\delta/2)$. Conversely, the quantum curve can be made to look classical by using $\phi = \pi \sin^2(\theta/2)$. Pointwise, the trick works. But the price is that the new

parameter is no longer the same physical knob. In general $\delta(\theta_1 + \theta_2) \neq \delta(\theta_1) + \delta(\theta_2)$, so the linear addition law of the original angle has been lost [3]. The reparameterization is not a homomorphism of the rotation-parameter group and does not intertwine the physical rotation action. The cheat therefore does not merely relabel probabilities; it changes the calibration through which the apparatus is varied.

Softmax belongs to the same discussion, although its natural knob is often a score, utility, or inverse-temperature parameter rather than a literal angle. It is included here only as an abstract response law into a simplex, not as a physical analogue of Born's quadratic amplitude rule. Given scores $u_C(a)$ and a strictly positive, usually monotone, link g , it forms

$$P_{g,C}(a) = \frac{g(u_C(a))}{\sum_{b \in C} g(u_C(b))}. \quad (6)$$

For $g_\beta(u) = \exp(\beta u)$, the parameter β interpolates between an almost uniform response at small β and a winner-take-all threshold as $\beta \rightarrow \infty$. This is useful as a link-function deformation, not as a claim that the exponential family has the same physical origin as Born's rule. Changing g , or changing the score scale, changes the response sensitivity of the model. For a single context this again guarantees only local normalization; it does not by itself specify which score scale is physically proper. If the same atom occurs in several contexts, admissibility adds a separate requirement: the response must be single-valued on that atom. Thus a softmax fit, a Malus curve, a linear classical curve, and a Heaviside step are not competing physical explanations. They are different response functions tied to different assumptions about what is being varied and which structure that variation is required to preserve.

Inverse knobs.— For a single monotone response curve this observation can be made completely explicit. Let $R_A(\theta)$ and $R_B(\theta)$ be two response functions, written here as equal-outcome probabilities. Equivalently one may use the correlation $E(\theta) = 2R(\theta) - 1$. If R_A is invertible on the range of R_B , then

$$T_{A \leftarrow B}(\theta) = R_A^{-1}(R_B(\theta)) \quad (7)$$

is the transcribed knob: feeding $T_{A \leftarrow B}(\theta)$ into response law A makes it reproduce response law B as a function of the displayed parameter θ . The injectivity assumption is substantive. Without a single-valued inverse one obtains at best a set-valued transcription or a limiting prescription, as in a threshold response.

For the classical linear response $R_{cl}(\theta) = \theta/\pi$ and the quantum singlet response $R_{qm}(\theta) = \sin^2(\theta/2)$, this gives

$$T_{cl \leftarrow qm}(\theta) = \pi \sin^2 \frac{\theta}{2}, \quad T_{qm \leftarrow cl}(\theta) = 2 \arcsin \sqrt{\frac{\theta}{\pi}}. \quad (8)$$

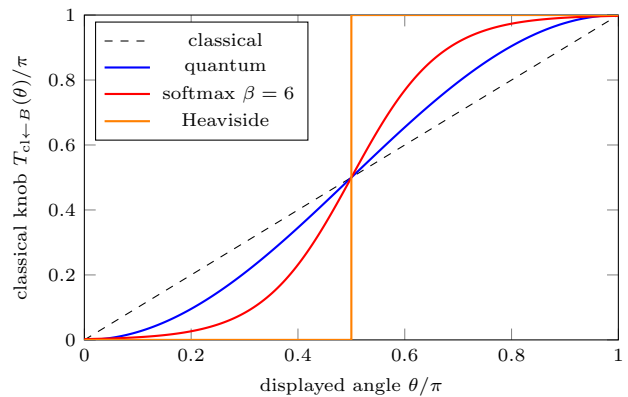


FIG. 1. Classical inverse knobs for several target response laws. Since $R_{cl}(\tau) = \tau/\pi$, the transcribed classical knob is simply $T_{cl \leftarrow B}(\theta)/\pi = R_B(\theta)$. The curves show how a linear classical response must be read on a generally non-symmetry-preserving scale in order to mimic a quantum, softmax, or Heaviside response as a function of the displayed angle.

These are precisely the parameter cheats: the first makes a classical linear response look quantum; the second makes the quantum response look classical.

For a centered binary softmax response

$$R_{\sigma,\beta}(\theta) = \{1 + \exp[-\beta(2\theta/\pi - 1)]\}^{-1}, \quad (9)$$

one obtains

$$T_{cl \leftarrow \sigma}(\theta) = \pi R_{\sigma,\beta}(\theta), \quad (10)$$

$$T_{\sigma \leftarrow cl}(\theta) = \frac{\pi}{2} \left[1 + \frac{1}{\beta} \log \frac{\theta/\pi}{1 - \theta/\pi} \right], \quad (11)$$

where the second formula is finite only for $0 < \theta/\pi < 1$, and it stays inside the physical interval only on the finite range actually covered by the chosen β . Similarly,

$$T_{qm \leftarrow \sigma}(\theta) = 2 \arcsin \sqrt{R_{\sigma,\beta}(\theta)}$$

and $T_{\sigma \leftarrow qm}$ is obtained by inserting $\sin^2(\theta/2)$ into the logit expression in Eq. (11). The Heaviside response $R_H(\theta) = H(2\theta/\pi - 1)$ can be transcribed into the classical knob as $T_{cl \leftarrow H}(\theta) = \pi R_H(\theta)$, but it has no ordinary inverse: an interval of angles is collapsed to one of two values. Its inverse is set-valued, or else appears only as a limiting threshold.

This single-curve trick should not be confused with a solution of a Bell or Boole problem. In a Bell–Clauser–Horne–Shimony–Holt (CHSH) experiment one does not merely fit one function $E(\theta)$. One must assign probabilities to several settings at once, such as a, b, a', b' , with one common underlying distribution over predetermined answers. A separate inverse knob can make each pairwise correlation look right, but those pairwise deformations need not come from one global angle scale. The failure

is already visible in the cheat parameters: $T(\theta_1 + \theta_2)$ generally differs from $T(\theta_1) + T(\theta_2)$. Thus the deformed pairwise angles need not be compatible with one geometry of four settings. The marginal-extension problem then reappears, and Farkas' lemma supplies the relevant obstruction.

The elastic-band examples make this warning concrete. Aerts' spin machine and its extended-Bloch descendants use a hidden break point, or more generally a hidden measurement interaction, to generate Born-like responses from an ordinary macroscopic mechanism [4–6]. The rubber-band analysis of context communication stresses the complementary limitation: recovering a cosine-shaped pair correlation in one adapted arrangement is not yet the same as a Bell-local simulation of a CHSH family [7]. If the band or share has to be realigned when the selected pair $(a, b), (a, b'), (a', b), (a', b')$ changes, then the knob transcription has become context communication. The shadow can be made to look quantum after breakfast, lunch, teatime, and supper, but the four shadows no longer arise from one setting-independent joint valuation. In Sven Aerts' Popescu–Rohrlich (PR)-box rubber-band construction the same point appears in a more extreme form: the resource is a single extended object whose breaking co-creates the two outcomes [8]. These models are therefore relevant here not as refutations of Boole–Bell, but as physical parables of where the response-function program stops: a single curve can be reparametrized, whereas a Bell web requires one common nonadaptive geometry or else a genuinely nonlocal or nonseparable resource.

Two contexts.— The same logic explains why two contexts, even when they share outcomes, are not yet enough to force a probability-level obstruction. Let

$$A = S \cup X, \quad B = S \cup Y \quad (12)$$

be two Boolean blocks, with shared atoms S , atoms X belonging only to A , and atoms Y belonging only to B . Suppose context-wise probabilities p^A and p^B are normalized, nonnegative, and agree on the overlap:

$$p_s^A = p_s^B =: r_s \quad (s \in S). \quad (13)$$

Let $m = 1 - \sum_{s \in S} r_s$. Then

$$\sum_{x \in X} p_x^A = m = \sum_{y \in Y} p_y^B. \quad (14)$$

If $m > 0$, choose for example

$$w_{xy} = \frac{p_x^A p_y^B}{m}; \quad (15)$$

if $m = 0$, set all $w_{xy} = 0$. Now define a classical sample space with one point λ_s of weight r_s for each shared atom s , and one point λ_{xy} of weight w_{xy} for each pair $x \in X, y \in Y$. At λ_s , the shared atom s is the true answer

in both contexts. At λ_{xy} , the answer to A is x and the answer to B is y . Reading the A -coordinate gives p^A , and reading the B -coordinate gives p^B .

Proposition 2. *Every pair of normalized context distributions satisfying Eq. (13) has a classical joint model over deterministic answers for the two blocks.*

This is just the elementary coupling theorem for two marginal distributions, with the shared outcomes treated as already coupled. Equivalently, after the common mass has been fixed, each residual cell obeys the finite Frechet–Hoeffding bounds

$$\max(0, p_x^A + p_y^B - m) \leq w_{xy} \leq \min(p_x^A, p_y^B), \quad (16)$$

and Eq. (15) is just one point in this coupling polytope. Thus two-context intertwining has no nontrivial marginal obstruction at the level of static weights. The response functions realizing those weights may still be very different: continuous or discontinuous, affine or Malus-type, softmax-shaped or boundary-valued. The obstruction requires a sufficiently constrained web of local shadows, typically a cycle or a Kochen–Specker-type pasting, so that all pairwise or context-wise agreements cannot be realized by one global nonnegative distribution.

Farkas alternative.— The extension problem can be written in the form

$$Mq = b, \quad q \geq 0, \quad (17)$$

where q is a distribution over deterministic global assignments and b collects the observed context probabilities. Farkas' lemma says that exactly one of the following alternatives holds:

$$\exists q \geq 0 \text{ such that } Mq = b, \quad (18)$$

$$\exists y \text{ such that } y^T M \geq 0 \text{ and } y^T b < 0. \quad (19)$$

The second line is a separating inequality. Garg and Mermin used precisely this logic to formulate the question of whether specified pair distributions can arise as marginals of compatible higher-order distributions [9]. Fine's theorem for Bell inequalities is a celebrated special case of the same joint-distribution viewpoint [10]; Pitowsky's correlation polytopes place the construction in convex-geometric form [11].

Equation (17) is also the finite-dimensional skeleton of simplex embeddability tests. A prepare-and-measure fragment is classical when its probability table factors through a simplex, or equivalently through a nonnegative cone generated by deterministic responses. Vectorizing the nonnegative factorization again yields a linear feasibility problem; infeasibility has a Farkas dual certificate. The recent generalized-probabilistic simplex criteria and their linear-programming implementations make this operational version explicit [12, 13].

Simplex and theta bodies.— For a fixed exclusivity structure there are two natural convex bodies. The classical body is generated by deterministic two-valued states:

$$\mathcal{P}_{\text{cl}} = \text{conv}\{\text{admissible 0/1 assignments}\}. \quad (20)$$

For graphs this is the stable-set polytope, denoted $\text{STAB}(G)$, possibly with additional normalization equations for complete contexts. One standard realization of the corresponding quantum body is obtained from squared projections of a state onto an orthogonal representation:

$$p_v = |\langle \psi | v \rangle|^2, \quad (21)$$

or by the corresponding density-operator formula. In graph language this is related to Lovasz-type theta bodies [14–16]. The larger admissible-weight body, often called the no-disturbance or consistent-exclusivity polytope in contextuality language, is obtained by keeping only non-negativity, context-wise normalization, additivity on exclusive events, and equality of values assigned to the same atom in different contexts.

For one clique, these bodies have the same operational shadow: the simplex. For two cliques pasted along a common clique, the explicit coupling above still keeps the classical and quantum context marginals aligned. Cycles change the situation. The pentagon is the minimal calibration. Let

$$C_i = \{a_i, a_{i+1}, x_i\}, \quad i = 1, \dots, 5,$$

with indices modulo 5. The cyclic atoms a_i are the intertwining observables, and the atoms x_i are context-specific. The assignment

$$p(a_i) = \frac{1}{2}, \quad p(x_i) = 0 \quad (i = 1, \dots, 5) \quad (22)$$

obeys completeness in every context, $p(a_i) + p(a_{i+1}) + p(x_i) = 1$, is additive on mutually exclusive events inside each Boolean block, and is atom-consistent: the same intertwining atom has the same value wherever it occurs. This is deliberately weaker than noncontextual hidden-variable embeddability in a classical simplex, and it is not Spekkens-style operational noncontextuality. Indeed, it is not a convex mixture of deterministic two-valued states.

Indeed, the classical noncontextual bound on the five cyclic events is

$$\sum_{i=1}^5 p_i \leq 2, \quad (23)$$

for the Klyachko–Can–Binicioğlu–Shumovsky (KCBS) pentagon inequality [17–19], whereas a faithful quantum orthogonal representation reaches

$$\sum_{i=1}^5 p_i = \sqrt{5}. \quad (24)$$

This is the Lovasz number of the pentagon exclusivity graph [16, 20]. The half-weight (22) gives $\sum_i p(a_i) = 5/2$, so it is admissible and atom-consistent, yet lies beyond both the classical bound and the usual KCBS/Lovasz quantum value. This is the KCBS/Lovasz-theta gap in its most economical form. In the language used here, the Hilbert-space squared-projection body has escaped the classical simplex body only after enough context-wise shadows have been required to glue together; the exotic half-weight escapes still farther inside the larger admissible/no-disturbance polytope. It is therefore best read as a boundary object of the admissible theory, not as a Hilbert-space-realizable quantum state.

For the pentagon this is also an explicit Farkas certificate. Deterministic classical assignments are the stable sets of C_5 , and each stable set contains at most two cyclic atoms. Hence the row vector $y = (1, 1, 1, 1, 1)$ separates the half-weight from the classical stable-set polytope:

$$y^T b = \frac{5}{2} \quad \text{whereas} \quad \max_{\chi \in \text{STAB}(C_5)} y^T \chi = 2. \quad (25)$$

Thus the KCBS inequality is precisely the Farkas dual witness for the nonexistence of a nonnegative convex combination of deterministic valuations realizing $b = (1/2, \dots, 1/2)$.

The response-function point is sharpest in this example. The half-weight may be approached by a generalized softmax path

$$p_r(a_i) = \frac{1}{2+r}, \quad p_r(x_i) = \frac{r}{2+r}, \quad (26)$$

where $r = g(t)/g(u_0)$. The uniform state occurs at $r = 1$, and the exotic half-weight is the boundary limit $r \rightarrow 0$. For an exponential softmax this is a limiting response rather than a finite-score response; for a Heaviside link one instead obtains a discontinuous knob response. The admissibility equations are the same in all cases, but the response function—the behavior of the knob—is very different.

Informal summary.— The moral is not that quantum probability is secretly classical, nor that a single probability vector contains enough information to reveal its origin. The moral is more local: one must keep track of what has been thrown away. When the generating geometry is forgotten, a single context leaves only a shadow. When the physical knob is retained, the shadow begins to move, and its calibrated, symmetry-constrained motion can already distinguish affine classical response from Malus-type Hilbert-space response or from discontinuous threshold behavior.

Two weakly pasted shadows still need not betray anything nonclassical. If two contexts agree on their shared atoms, the remaining mass can be coupled by an ordinary joint distribution. The obstruction appears only when enough local shadows are required to hang together as

one global object. Then the question becomes a linear feasibility problem: either there is a classical joint model or simplex factorization, or Farkas' lemma supplies the separating inequality. The rubber-band models are the laboratory parable: with context communication, adaptive realignment, or one extended nonseparable band, they can produce quantum-looking shadows, but not a single Bell-local valuation glued across all four settings.

Thus the staged picture is: shadow, calibrated knob, weak pasting, global web. The frozen shadow hides classical, quantum, and exotic origins. The knob records how the shadow changes under symmetry-constrained variation. Weak pastings remain extendable. Rich webs can fail to extend. The pentagon half-weight finally reminds us that admissibility and atom-consistency are still not the same as classical embeddability or as a physical response law: the same static constraints can be obeyed by very different knobs.

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